National Aeronautics and
Space Administration





NASA SYSTEMS ENGINEERING HANDBOOK





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scope and authority of the forum and identify necessary voting or nonvoting participants. Ad hoc members may be identified when the expertise or input of specific stakeholders is needed when specific topics are addressed. It is important to ensure that stakeholders have been identified to support the forum.

6.1.1.2.6 *Issue Technical Work Directives*

The technical team provides technical work directives to Cost Account Managers (CAMs). This enables the CAMs to prepare detailed plans that are mutually consistent and collectively address all of the work to be performed. These plans include the detailed schedules and budgets for cost accounts that are needed for cost management and EVM.

Issuing technical work directives is an essential activity during Phase B of a project when a detailed planning baseline is required. If this activity is not implemented, then the CAMs are often left with insufficient guidance for detailed planning. The schedules and budgets that are needed for EVM will then be based on assumptions and local interpretations of project-level information. If this is the case, it is highly likely that substantial variances will occur between the baseline plan and the work performed. Providing technical work directives to CAMs produces a more organized technical team. This activity may be repeated when replanning occurs.

This "technical work directives" step produces: (1) planning directives to CAMs that result in (2) a consistent set of cost account plans. Where EVM is called for, it produces (3) an EVM planning baseline, including a Budgeted Cost of Work Scheduled (BCWS).

This activity is not limited to systems engineering. This is a normal part of project planning wherever there is a need for an accurate planning baseline. For additional information on Technical Work Directives, refer to Section 6.1.1.2.6 in the NASA Expanded Guidance for Systems Engineering at *https://nen.nasa.gov/web/se/doc-repository*.

6.1.1.2.7 *Capture Technical Planning Work Products*

The work products from the Technical Planning Process should be managed using either the Technical Data Management Process or the Configuration Management Process as required. Some of the more important products of technical planning (i.e., the WBS, the SEMP, and the schedule, etc.) are kept under configuration control and captured using the CM process. The Technical Data Management Process is used to capture trade studies, cost estimates, technical analyses, reports, and other important documents not under formal configuration control. Work products, such as meeting minutes and correspondence (including e-mail) containing decisions or agreements with stakeholders also should be retained and stored in project files for later reference.

6.1.1.3 *Outputs*

Typical outputs from technical planning activities are:

- **Technical work cost estimates, schedules, and resource needs:** e.g., funds, workforce, facilities, and equipment (to the project) within the project resources;
- **Product and process measures:** Those needed to assess progress of the technical effort and the effectiveness of processes (to the Technical Assessment Process);
- **SEMP and other technical plans:** Technical planning strategy, WBS, SEMP, HSI Plan, V&V Plan, and other technical plans that support implementation of the technical effort (to all processes; applicable plans to technical processes);

- **Technical work directives:** e.g., work packages or task orders with work authorization (to applicable technical teams); and
- **Technical Planning Process work products:** Includes products needed to provide reports, records, and non-deliverable outcomes of process activities (to the Technical Data Management Process).

The resulting technical planning strategy constitutes an outline, or rough draft, of the SEMP. This serves as a starting point for the overall Technical Planning Process after initial preparation is complete. When preparations for technical planning are complete, the technical team should have a cost estimate and schedule for the technical planning effort. The budget and schedule to support the defined technical planning effort can then be negotiated with the project manager to resolve any discrepancies between what is needed and what is available. The SEMP baseline needs to be completed. Planning for the update of the SEMP based on programmatic changes needs to be developed and implemented. The SEMP needs to be approved by the appropriate level of authority.

6.1.2 Technical Planning Guidance

Refer to Section 6.1.2 in the NASA Expanded Guidance for Systems Engineering at *https://nen.nasa.gov/web/se/doc-repository* for additional guidance on:

- Work Breakdown Structure (WBS),
- cost definition and modeling, and
- lessons learned.

Additional information on the WBS can also be found in *NASA/SP-2010-3404, NASA Work Breakdown Structure Handbook* and on costing in the *NASA Cost Estimating Handbook.*

6.2 Requirements Management

Requirements management activities apply to the management of all stakeholder expectations, customer requirements, and technical product requirements down to the lowest level product component requirements (hereafter referred to as expectations and requirements). This includes physical functional and operational requirements, including those that result from interfaces between the systems in question and other external entities and environments. The Requirements Management Process is used to:

- Identify, control, decompose, and allocate requirements across all levels of the WBS.
- Provide bidirectional traceability.

DEFINITIONS

Traceability: A discernible association between two or more logical entities such as requirements, system elements, verifications, or tasks.

Bidirectional traceability: The ability to trace any given requirement/expectation to its parent requirement/expectation and to its allocated children requirements/expectations.

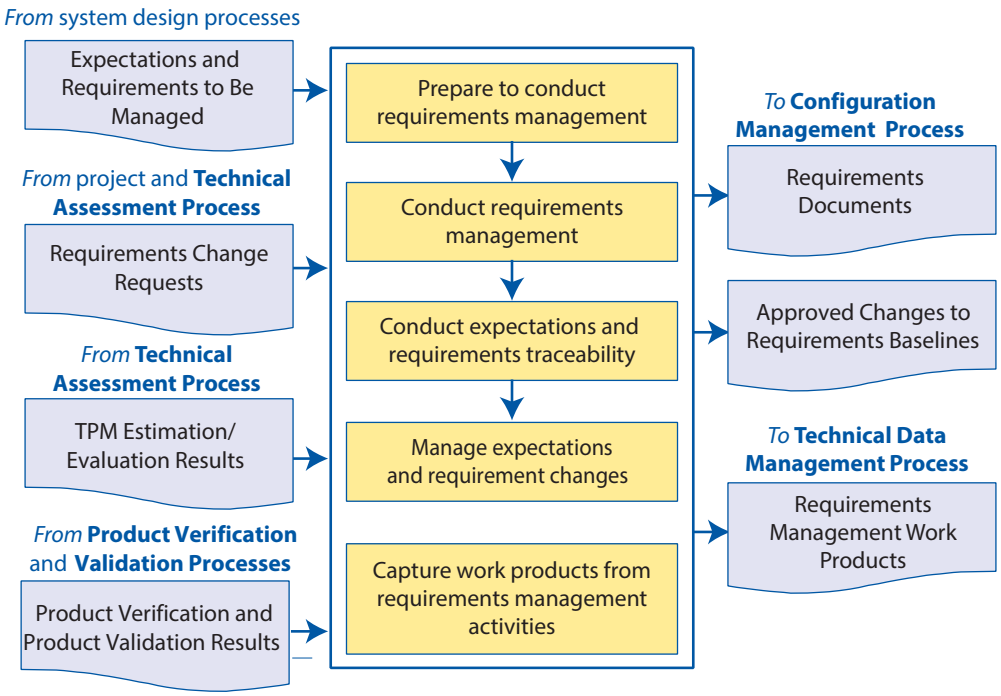


FIGURE 6.2-1 Requirements Management Process

- Manage the changes to established requirement baselines over the life cycle of the system products.

6.2.1 Process Description

FIGURE 6.2-1 provides a typical flow diagram for the Requirements Management Process and identifies typical inputs, outputs, and activities to consider in addressing requirements management.

6.2.1.1 *Inputs*

There are several fundamental inputs to the Requirements Management Process.

- **Expectations and requirements to be managed:** Requirements and stakeholder expectations are identified during the system design processes, primarily from the Stakeholder Expectations Definition Process and the Technical Requirements Definition Process.
- **Requirement change requests:** The Requirements Management Process should be prepared to deal with requirement change requests that can be generated at any time during the project life cycle or as a result of reviews and assessments as part of the Technical Assessment Process.
- **TPM estimation/evaluation results:** TPM estimation/evaluation results from the Technical Assessment Process provide an early warning of the adequacy of a design in satisfying selected critical technical parameter requirements. Variances from expected values of product performance may trigger changes to requirements.
- **Product verification and validation results:** Product verification and product validation results from the Product Verification and Product Validation Processes are mapped into the

requirements database with the goal of verifying and validating all requirements.

6.2.1.2 Process Activities

6.2.1.2.1 *Prepare to Conduct Requirements Management*

Preparing to conduct requirements management includes gathering the requirements that were defined and baselined during the Requirements Definition Process. Identification of the sources/owners of each requirement should be checked for currency. The organization (e.g., change board) and procedures to perform requirements management are established.

6.2.1.2.2 *Conduct Requirements Management*

The Requirements Management Process involves managing all changes to expectations and requirements baselines over the life of the product and maintaining bidirectional traceability between stakeholder expectations, customer requirements, technical product requirements, product component requirements, design documents, and test plans and procedures. The successful management of requirements involves several key activities:

- Establish a plan for executing requirements management.
- Receive requirements from the system design processes and organize them in a hierarchical tree structure.
- Maintain bidirectional traceability between requirements.
- Evaluate all change requests to the requirements baseline over the life of the project and make changes if approved by change board.
- Maintain consistency between the requirements, the ConOps, and the architecture/design, and initiate corrective actions to eliminate inconsistencies.

6.2.1.2.3 *Conduct Expectations and Requirements Traceability*

As each requirement is documented, its bidirectional traceability should be recorded. Each requirement should be traced back to a parent/source requirement or expectation in a baselined document or identified as *self-derived* and concurrence on it sought from the next higher level requirements sources. Examples of self-derived requirements are requirements that are locally adopted as good practices or are the result of design decisions made while performing the activities of the Logical Decomposition and Design Solution Processes.

The requirements should be evaluated, independently if possible, to ensure that the requirements trace is correct and that it fully addresses its parent requirements. If it does not, some other requirement(s) should complete fulfillment of the parent requirement and be included in the traceability matrix. In addition, ensure that all top-level parent document requirements have been allocated to the lower level requirements. If there is no parent for a particular requirement and it is not an acceptable self-derived requirement, it should be assumed either that the traceability process is flawed and should be redone or that the requirement is "gold plating" and should be eliminated. Duplication between levels should be resolved. If a requirement is simply repeated at a lower level and it is not an externally imposed constraint, it may not belong at the higher level. Requirements traceability is usually recorded in a requirements matrix or through the use of a requirements modeling application.

6.2.1.2.4 *Managing Expectations and Requirement Changes*

Throughout early Phase A, changes in requirements and constraints will occur as they are initially defined and matured. It is imperative that all changes be thoroughly evaluated to determine the impacts on the cost, schedule, architecture, design, interfaces, ConOps, and higher and lower level requirements. Performing functional and sensitivity analyses will ensure that the requirements are realistic and evenly allocated. Rigorous requirements verification and validation will ensure that the requirements can be satisfied and conform to mission objectives. All changes should be subjected to a review and approval cycle to maintain traceability and to ensure that the impacts are fully assessed for all parts of the system.

Once the requirements have been validated and reviewed in the System Requirements Review (SRR) in late Phase A, they are placed under formal configuration control. Thereafter, any changes to the requirements should be approved by a Configuration Control Board (CCB) or equivalent authority. The systems engineer, project manager, and other key engineers usually participate in the CCB approval processes to assess the impact of the change including cost, performance, programmatic, and safety.

Requirement changes during Phases B and C are more likely to cause significant adverse impacts to the project cost and schedule. It is even more important that these late changes are carefully evaluated to fully understand their impact on cost, schedule, and technical designs.

The technical team should also ensure that the approved requirements are communicated in a timely manner to all relevant people. Each project should have already established the mechanism to track and disseminate the latest project information. Further information on Configuration Management (CM) can be found in *Section 6.5*.

6.2.1.2.5 *Key Issues for Requirements Management*

Requirements Changes

Effective management of requirements changes requires a process that assesses the impact of the proposed changes prior to approval and implementation of the change. This is normally accomplished through the use of the Configuration Management Process. In order for CM to perform this function, a baseline configuration should be documented and tools used to assess impacts to the baseline. Typical tools used to analyze the change impacts are as follows:

- **Performance Margins:** This tool is a list of key performance margins for the system and the current status of the margin. For example, the propellant performance margin will provide the necessary propellant available versus the propellant necessary to complete the mission. Changes should be assessed for their impact on performance margins.

- **CM Topic Evaluators List:** This list is developed by the project office to ensure that the appropriate persons are evaluating the changes and providing impacts to the change. All changes need to be routed to the appropriate individuals to ensure that the change has had all impacts identified. This list will need to be updated periodically.

- **Risk System and Threats List:** The risk system can be used to identify risks to the project and the cost, schedule, and technical aspects of the risk. Changes to the baseline can affect the consequences and likelihood of identified risk or can introduce new risk to the project. A threats list is normally used to identify the costs associated with all the risks for the project. Project reserves are used to mitigate the appropriate risk. Analyses of the reserves available versus the needs identified by the threats list assist in the prioritization for reserve use.

The process for managing requirements changes needs to take into account the distribution of information related to the decisions made during the change process. The Configuration Management Process needs to communicate the requirements change decisions to the affected organizations. During a board meeting to approve a change, actions to update documentation need to be included as part of the change package. These actions should be tracked to ensure that affected documentation is updated in a timely manner.

Requirements Creep

"Requirements creep" is the term used to describe the subtle way that requirements grow imperceptibly during the course of a project. The tendency for the set of requirements is to relentlessly increase in size during the course of development, resulting in a system that is more expensive and complex than originally intended. Often the changes are quite innocent and what appear to be changes to a system are really enhancements in disguise.

However, some of the requirements creep involves truly new requirements that did not exist, and could not have been anticipated, during the Technical Requirements Definition Process. These new requirements are the result of evolution, and if we are to build a relevant system, we cannot ignore them.

There are several techniques for avoiding or at least minimizing requirements creep:

- The first line of defense is a good ConOps that has been thoroughly discussed and agreed-to by the customer and relevant stakeholders.
- In the early requirements definition phase, flush out the conscious, unconscious, and undreamed-of requirements that might otherwise not be stated.
- Establish a strict process for assessing requirement changes as part of the Configuration Management Process.
- Establish official channels for submitting change requests. This will determine who has the authority to generate requirement changes and submit them formally to the CCB (e.g., a contractor-designated representative, project technical leads, customer/science team lead, or user).
- Measure the functionality of each requirement change request and assess its impact on the rest of the system. Compare this impact with the consequences of not approving the change. What is the risk if the change is not approved?
- Determine if the proposed change can be accommodated within the fiscal and technical resource budgets. If it cannot be accommodated within the established resource margins, then the change most likely should be denied.

6.2.1.2.6 *Capture Work Products*

These products include maintaining and reporting information on the rationale for and disposition and implementation of change actions, current requirement compliance status and expectation, and requirement baselines.

6.2.1.3 *Outputs*

Typical outputs from the requirements management activities are:

- **Requirements Documents:** Requirements documents are submitted to the Configuration Management Process when the requirements are baselined. The official controlled versions of these documents are generally maintained in electronic format within the requirements management tool

that has been selected by the project. In this way, they are linked to the requirements matrix with all of its traceable relationships.

- **Approved Changes to the Requirements Baselines:** Approved changes to the requirements baselines are issued as an output of the Requirements Management Process after careful assessment of all the impacts of the requirements change across the entire product or system. A single change can have a far-reaching ripple effect, which may result in several requirement changes in a number of documents.

- **Various Requirements Management Work Products:** Requirements management work products are any reports, records, and undeliverable outcomes of the Requirements Management Process. For example, the bidirectional traceability status would be one of the work products that would be used in the verification and validation reports.

6.2.2 Requirements Management Guidance

Refer to Section 6.2.2 in the NASA Expanded Guidance for Systems Engineering at *https://nen.nasa.gov/web/se/doc-repository* for additional guidance on:

- the Requirements Management Plan and
- requirements management tools.

6.3 Interface Management

The definition, management, and control of interfaces are crucial to successful programs or projects. Interface management is a process to assist in controlling product development when efforts are divided among parties (e.g., Government, contractors, geographically diverse technical teams, etc.) and/or to define and maintain compliance among the products that should interoperate.

The basic tasks that need to be established involve the management of internal and external interfaces of the various levels of products and operator tasks to support product integration. These basic tasks are as follows:

- Define interfaces;

- Identify the characteristics of the interfaces (physical, electrical, mechanical, human, etc.);

- Ensure interface compatibility at all defined interfaces by using a process documented and approved by the project;

- Strictly control all of the interface processes during design, construction, operation, etc.;

- Identify lower level products to be assembled and integrated (from the Product Transition Process);

- Identify assembly drawings or other documentation that show the complete configuration of the product being integrated, a parts list, and any assembly instructions (e.g., torque requirements for fasteners);

- Identify end-product, design-definition-specified requirements (specifications), and configuration documentation for the applicable work breakdown structure model, including interface specifications, in the form appropriate to satisfy the product life cycle phase success criteria (from the Configuration Management Process); and

- Identify product integration-enabling products (from existing resources or the Product Transition Process for enabling product realization).

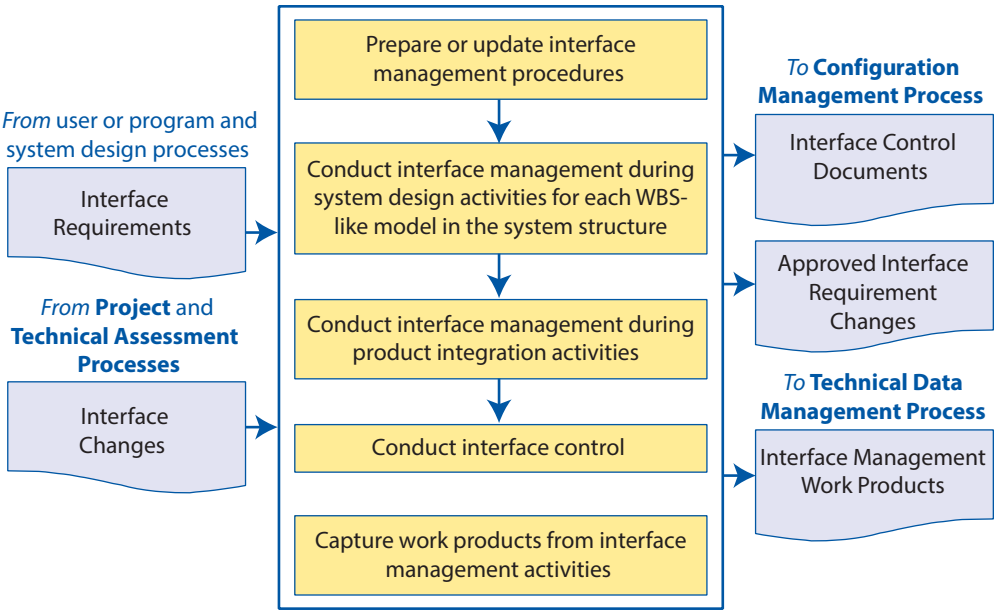


FIGURE 6.3-1 Interface Management Process

6.3.1 Process Description

FIGURE 6.3-1 provides a typical flow diagram for the Interface Management Process and identifies typical inputs, outputs, and activities to consider in addressing interface management.

6.3.1.1 *Inputs*

Typical inputs needed to understand and address interface management would include the following:

- **Interface Requirements:** These include the internal and external functional, physical, and performance interface requirements developed as part of the Technical Requirements Definition Process for the product(s).
- **Interface Change Requests:** These include changes resulting from program or project agreements or changes on the part of the technical team as part of the Technical Assessment Process.

Other inputs that might be useful are:

- **System Description:** This allows the design of the system to be explored and examined to determine where system interfaces exist. Contractor arrangements will also dictate where interfaces are needed.
- **System Boundaries:** Documented physical boundaries, components, and/or subsystems, which are all drivers for determining where interfaces exist.
- **Organizational Structure:** Decisions on which organization will dictate interfaces, particularly when there is the need to jointly agree on shared interface parameters of a system. The program and project WBS will also provide organizational interface boundaries.
- **Boards Structure:** Defined board structure that identifies organizational interface responsibilities.

6.3.1.2 Process Activities

6.3.1.2.1 *Prepare or Update Interface Management Procedures*

These procedures establish the interface management responsibilities, what process will be used to maintain and control the internal and external functional and physical interfaces (including human), and how the change process will be conducted. Training of the technical teams or other support may also be required and planned.

6.3.1.2.2 *Conduct Interface Management during System Design Activities*

During project Formulation, the ConOps of the product is analyzed to identify both external and internal interfaces. This analysis will establish the origin, destination, stimuli, and special characteristics of the interfaces that need to be documented and maintained. As the system structure and architecture emerges, interfaces will be added and existing interfaces will be changed and should be maintained. Thus, the Interface Management Process has a close relationship to other areas, such as requirements definition and configuration management, during this period.

6.3.1.2.3 *Conduct Interface Management during Product Integration*

During product integration, interface management activities would support the review of integration and assembly procedures to ensure interfaces are properly marked and compatible with specifications and interface control documents. The interface management process has a close relationship to verification and validation. Interface control documentation and approved interface requirement changes are used as inputs to the Product Verification Process and the Product Validation Process, particularly where verification test constraints and interface parameters are needed to set the test objectives and test plans. Interface requirements verification is a critical aspect of the overall system verification.

6.3.1.2.4 *Conduct Interface Control*

Typically, an Interface Working Group (IWG) establishes communication links between those responsible for interfacing systems, end products, enabling products, and subsystems. The IWG has the responsibility to ensure accomplishment of the planning, scheduling, and execution of all interface activities. An IWG is typically a technical team with appropriate technical membership from the interfacing parties (e.g., the project, the contractor, etc.). The IWG may work independently or as a part of a larger change control board.

6.3.1.2.5 *Capture Work Products*

Work products include the strategy and procedures for conducting interface management, rationale for interface decisions made, assumptions made in approving or denying an interface change, actions taken to correct identified interface anomalies, lessons learned and updated support and interface agreement documentation.

6.3.1.3 Outputs

Typical outputs needed to capture interface management would include:

- **Interface control documentation.** This is the documentation that identifies and captures the interface information and the approved interface change requests. Types of interface documentation include the Interface Requirements Document (IRD), Interface Control Document/Drawing (ICD), Interface Definition Document (IDD), and Interface Control Plan (ICP). These outputs will then be maintained and approved using the Configuration Management Process and become a part of the overall technical data package for the project.

- **Approved interface requirement changes.** After the interface requirements have been baselined, the Requirements Management Process should be used to identify the need for changes, evaluate the impact of the proposed change, document the final approval/disapproval, and update the requirements documentation/tool/database. For interfaces that require approval from all sides, unanimous approval is required. Changing interface requirements late in the design or implementation life cycle is more likely to have a significant impact on the cost, schedule, or technical design/operations.

- **Other work products.** These work products include the strategy and procedures for conducting interface management, the rationale for interface decisions made, the assumption made in approving or denying an interface change, the actions taken to correct identified interface anomalies, the lessons learned in performing the interface management activities, and the updated support and interface agreement documentation.

6.3.2 Interface Management Guidance

Refer to Section 6.3.2 in the NASA Expanded Guidance for Systems Engineering at *https://nen.nasa.gov/web/se/doc-repository* for additional guidance on:

- interface requirements documents,
- interface control documents,
- interface control drawings,
- interface definition documents,
- the interface control plans, and
- interface management tasks.

6.4 Technical Risk Management

The Technical Risk Management Process is one of the crosscutting technical management processes. Risk is the potential for performance shortfalls, which may be realized in the future, with respect to achieving explicitly established and stated performance requirements. The performance shortfalls may be related to institutional support for mission execution or related to any one or more of the following mission execution domains:

- safety
- technical
- cost
- schedule

Systems engineers are involved in this process to help identify potential technical risks, develop mitigation plans, monitor progress of the technical effort to determine if new risks arise or old risks can be retired, and to be available to answer questions and resolve issues. The following is guidance in implementation of risk management in general. Thus, when implementing risk management on any given program/project, the responsible systems engineer should direct the effort accordingly. This may involve more or less rigor and formality than that specified in governing documents such as NPRs. Of course, if deviating from NPR "requirements," the responsible engineer must follow the deviation approval process. The idea is to tailor the risk management process so that it meets the needs of the individual program/project being executed while working within the bounds of the governing documentation (e.g., NPRs). For detailed information on the Risk Management Process, refer to the *NASA Risk Management Handbook (NASA/SP-2011-3422).*

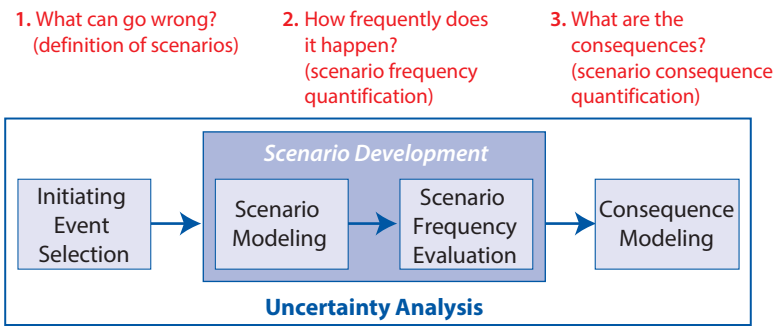


FIGURE 6.4-1 Risk Scenario Development (Source: NASA/SP-2011-3421)

Risk is characterized by three basic components:

1. The scenario(s) leading to degraded performance with respect to one or more performance measures (e.g., scenarios leading to injury, fatality, destruction of key assets; scenarios leading to exceedance of mass limits; scenarios leading to cost overruns; scenarios leading to schedule slippage);

2. The likelihood(s) (qualitative or quantitative) of those scenario(s); and

3. The consequence(s) (qualitative or quantitative severity of the performance degradation) that would result if the scenario(s) was (were) to occur.

Uncertainties are included in the evaluation of likelihoods and consequences.

Scenarios begin with a set of initiating events that cause the activity to depart from its intended state. For each initiating event, other events that are relevant to the evolution of the scenario may (or may not) occur and may have either a mitigating or exacerbating effect on the scenario progression. The frequencies of scenarios with undesired consequences are determined. Finally, the multitude of such scenarios is put together, with an understanding of the uncertainties, to create the risk profile of the system.

This "risk triplet" conceptualization of risk is illustrated in FIGURES 6.4-1 and 6.4-2.

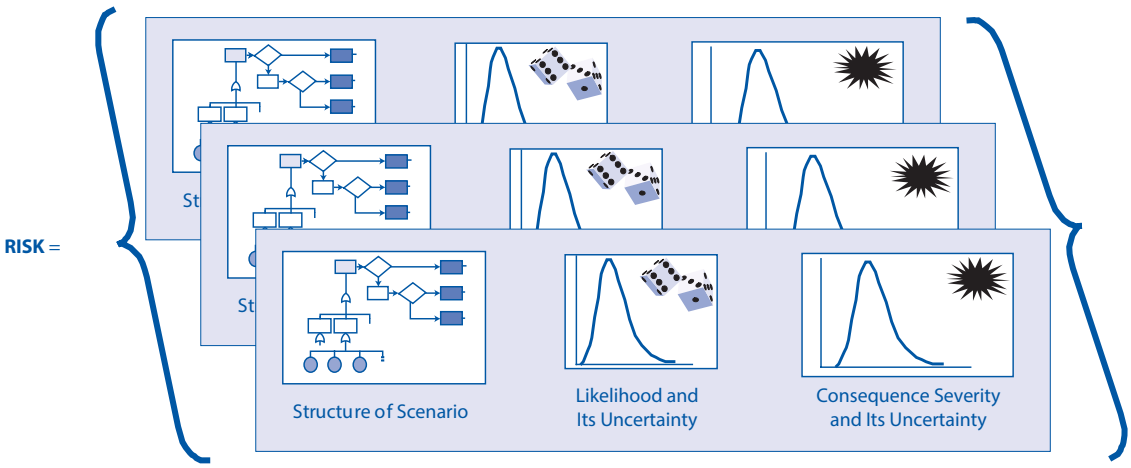


FIGURE 6.4-2 Risk as an Aggregate Set of Risk Triplets